System Overview and Architecture

Status date: April 29, 2026 **Audience:** the next developer who picks up Omnilore Scheduler from Team 34
Prerequisites: familiarity with Flutter/Dart and basic AWS EC2 concepts

1. Problem and Audience

Omnilore is a member-led lifelong-learning organization in Rancho Palos Verdes, CA. Every term, members rank course preferences (first choices and backups), declare their availability across twenty time slots, and request a specific number of classes (0–6). A small administrative team then has to compute a term schedule that:

1. drops courses that fall under a minimum class size,
2. splits oversize courses into two or more sections,
3. assigns each surviving course to one of twenty time slots,
4. assigns one main coordinator and a co-coordinator (or two equal co-coordinators) to each course, and
5. exposes four export actions: an early roster in phone-roster style, a final roster with coordinator labels, a mail-merge file, and an unmet-wants report.

The previous (pre-2026) implementation was a Flutter desktop application that depended on `dart:io`, native window sizing, and OS file dialogs. Onboarding a new admin meant installing Flutter and the desktop runtime. Several stakeholder-reported defects (the "Co-Coordinator" assignment bug and the "Show Splits" bug) and a missing intermediate-state save/load capability remained outstanding.

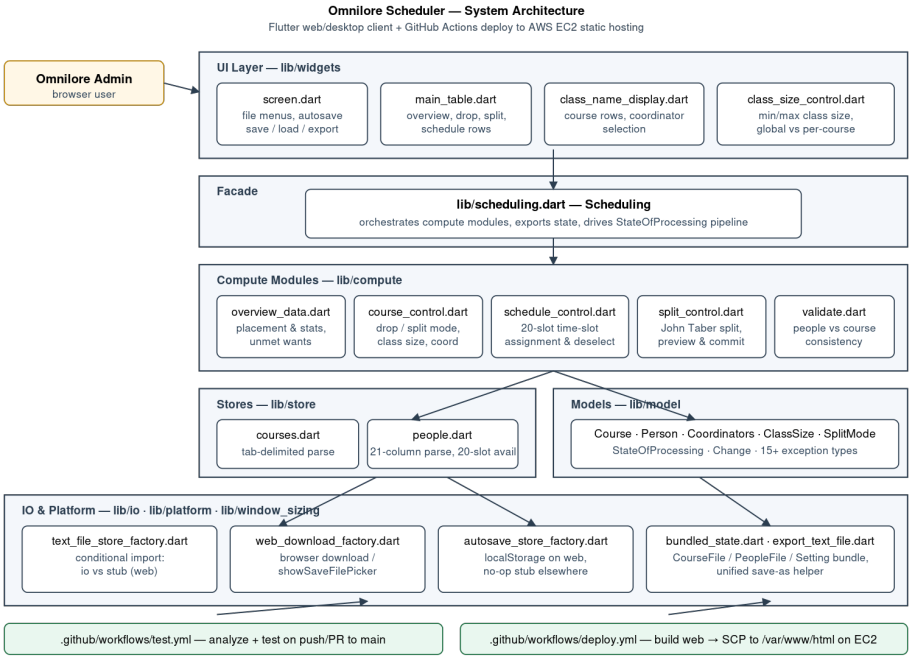
Team 34's mandate (CSCI 401, Spring 2026) was to migrate the application to a web-based delivery model, fix the two defects, add intermediate state save/load aligned with admin expectations, expand regression testing, and stand up automated cloud deployment. An optional AI-assisted scheduling feature was scoped out by the stakeholder in favor of stability.

2. Features Delivered (D1 → D7)

- Browser-deployable build. The same Dart codebase compiles for desktop and web. Conditional imports route file I/O, downloads, autosave, and window sizing to platform-appropriate implementations. A static check (scripts/check_platform_gating.sh) prevents dart:io and other desktop-only imports from leaking into shared code.
- Two-file load. Tab-delimited course and people text files are parsed in-browser via file_picker and validated against fifteen-plus exception types defined in lib/model/exceptions.dart.
- Drop/Split/Schedule/Coordinator pipeline. The original desktop pipeline is preserved end-to-end; oversize courses now expose a non-destructive Show Splits preview with manual rebalancing and explicit Cancel / Implement controls.
- Coordinator UX rebuild. Both Set C and CC and Set CC1 and CC2 now reset selection state on re-entry so admins can fully reassign coordinators; tapping a highlighted name on a coordinator-assigned course clears the assignment.
- Time-slot deselection. Clicking the same slot a second time unschedules the course; downstream availability counts recompute.
- Save / Save As / Load / Autosave / Restore. lib/io/bundled_state.dart produces a single text bundle containing the original course file, the original people file, and the scheduler state grammar — restoring everything from one file. Web autosave uses six localStorage keys with timestamped restore-prompts.

- Four export menu actions. Early roster (phone-roster style), final roster (with (C) / (CC) labels), tab-delimited mail-merge, and unmet-wants report. All exports route through a unified helper (lib/io/export_text_file.dart) with timestamped filenames from lib/io/default_filename.dart; web exports use showSaveFilePicker where the browser supports it and fall back to a normal browser download.
- Cloud deployment. .github/workflows/deploy.yml builds the Flutter web target, briefly whitelists the runner IP in an EC2 security group, copies build/web/* to /var/www/html over SCP, and revokes the IP — reachable today at http://scheduler.omnilore.org.
- Continuous validation. .github/workflows/test.yml runs the gating check, flutter analyze, and flutter test on every push and PR to main.

3. Architecture



A high-level view of how the layers fit together:

```
flowchart TD
    User["Omnilore admin (browser)"] --> UI["UI layer<br/>lib/widgets"]
    UI --> Facade["Scheduling facade<br/>lib/scheduling.dart"]
    Facade --> Compute["Compute modules<br/>lib/compute/*"]
    Facade --> Stores["Stores<br/>lib/store/*"]
    Facade --> Models["Models<br/>lib/model/*"]
    UI --> IO["I/O abstractions<br/>lib/io · lib/platform · lib/window_sizing"]
    IO --> Web["Browser: file_picker, showSaveFilePicker, localStorage"]
    IO --> Desktop["Desktop: dart:io paths, desktop_window"]
    GitHub["GitHub Actions"] --> CI["test.yml: gating + analyze + test"]
    GitHub --> Deploy["deploy.yml: build web → SCP to EC2"]
    Deploy --> EC2["AWS EC2<br>/var/www/html"]
    EC2 --> User
```

Layer responsibilities

Layer	Path	Job
UI	lib/widgets/	Owns Flutter state and renders the main table; hands user actions to Scheduling and reacts to its callback. The most active widget files are screen.dart, class_name_display.dart, class_size_control.dart, and widgets/table/*.
Facade	lib/scheduling.dart	The only object the UI talks to. Loads files, delegates compute to the modules below, computes StateOfProcessing after every change, and produces all exports. ~809 LOC.
Compute	lib/compute/	Pure scheduling logic (overview_data.dart, course_control.dart, schedule_control.dart, split_control.dart, validate.dart). No Flutter, no dart:io.
Models	lib/model/	Plain-Dart data carriers: Course, Person, Coordinators, ClassSize, SplitMode, Change, StateOfProcessing, exceptions.dart.
Stores	lib/store/	Tab-delimited parsers for the course and people files. Throw the exception types in lib/model/exceptions.dart on malformed input.
IO	lib/io/, lib/platform/, lib/window_sizing/	Conditional-import factories that pick a desktop or browser implementation at compile time.

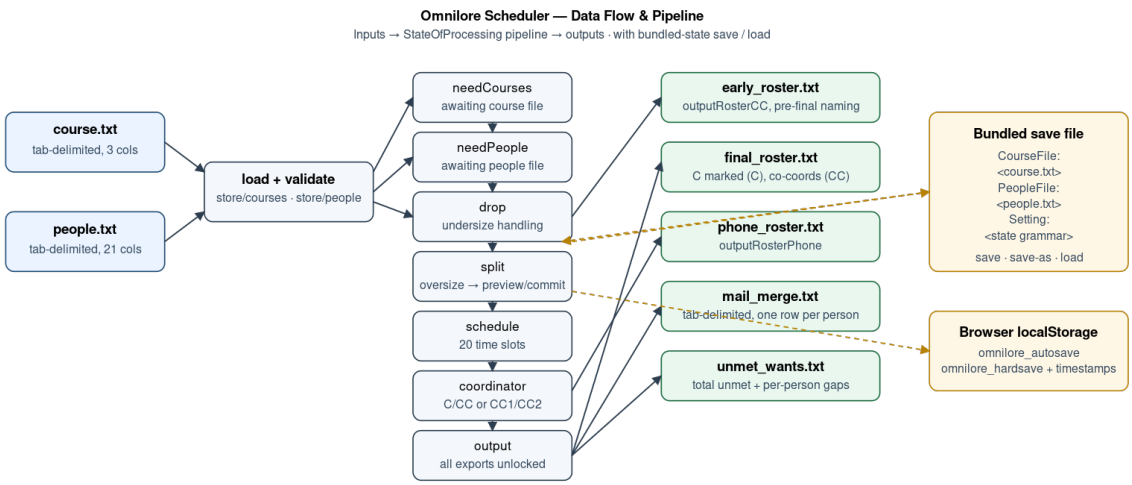
Conditional-import pattern (platform gating)

Every desktop-only API is hidden behind a three-file factory:

```
// text_file_store_factory.dart
export 'text_file_store_stub.dart'
    if (dart.library.io) 'package:omnilore_scheduler/io/text_file_store_io.dart';
```

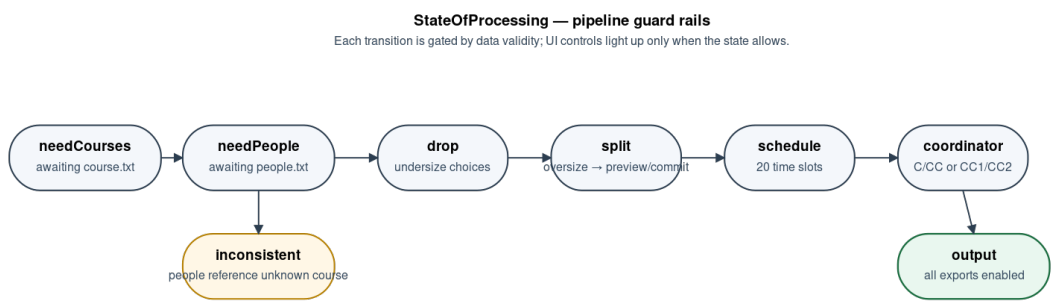
`scripts/check_platform_gating.sh` greps the entire `lib/` tree and **fails the build** if `import 'dart:io';` or `import 'package:desktop_window/...';` appears anywhere outside the explicit allowlist. This is also enforced by `test/platform_gating_test.dart`. Whenever you add a new platform-touching feature, follow the same triple of `*_factory.dart` + `*_stub.dart` + `*_io.dart` (or `*_web.dart`) and update the gating script.

4. Data Flow




```
flowchart LR
    Course["course.txt<br/>tab-delimited, 3 cols"] --> Stores
    People["people.txt<br/>tab-delimited, 21 cols"] --> Stores
    Stores --> Facade["Scheduling"]
    Facade --> Drop["drop"] --> Split["split"] --> Sched["schedule"] --> Coord["coordinator"] --> Out["output"]
    Out --> Early["early_roster_YYYY-MM-DD_HHMM.txt"]
    Out --> Roster["final_roster_YYYY-MM-DD_HHMM.txt"]
    Out --> MM["mail_merge_YYYY-MM-DD_HHMM.txt"]
    Out --> Unmet["unmet_wants_YYYY-MM-DD_HHMM.txt"]
    Facade --> Bundle["Bundled save:<br/>CourseFile + PeopleFile + Setting"]
    Bundle --> Auto["Browser localStorage<br/>(autosave + hardsave)"]
    Bundle --> File["User-saved .txt"]
```

5. Pipeline State Machine



`Scheduling._updateStateOfProcessing()` evaluates the eight `StateOfProcessing` values **in order** after every UI-driven `compute(Change)` call:

`needCourses → needPeople → inconsistent | drop | split | schedule | coordinator → output`

State	Entered when	UI hint
needCourses	course store is empty	"Load courses."
needPeople	course OK, people store is empty	"Load people."
inconsistent	a person references an unknown course code	Inline error; admin must fix the input file.
drop	at least one course is below its minimum class size	Drop / split-mode controls light up.

State	Entered when	UI hint
split	at least one course is over its maximum size	Show Splits preview is required.
schedule	all classes sized; some unscheduled	Twenty-slot table is live.
coordinator	all scheduled; coordinators incomplete	C/CC and CC1/CC2 controls light up.
output	everything assigned	All exports are unlocked.

The pipeline is **monotonic forward** but reactive backward: dropping a course or rebalancing a split rewinds downstream stages and the UI greys out their controls until the upstream constraint is satisfied again.

6. Key Design Choices

- Single Dart engine for desktop and web. The compute and model layers are pure Dart with no Flutter or dart:io dependencies. This keeps the desktop-vs-web fork to the I/O surface only, which is the smallest possible blast radius.
- Bundled save format. A single text file (CourseFile: + PeopleFile: + Setting: sections) restores the entire working session. This was chosen over JSON because (a) it remains diff-friendly and human-readable for stakeholder review, (b) it allows admins to hand-edit a snapshot if needed, and (c) it preserves the original tab-delimited input bytes — useful for traceability and re-running.
- Autosave is convenience, not durability. Web autosave writes to browser localStorage and is therefore tied to that browser profile and device. The user-facing message and the runbook treat Save As to a real file as the only operational backup.
- No backend. The application is entirely client-side. There is no database, no auth, no API, and no PII server-side. PII (names, phone numbers) lives only in the admin's browser session and in the files they choose to save.
- Static-asset deploy. The web build is just static files. EC2 + nginx/Apache was the path of least resistance because Omnilore already has an AWS account; the doc includes a recommendation to migrate to S3 + CloudFront in the long term (see Backlog_Known_Issues_Roadmap.md, P2).
- Aggressive platform gating. Catching a desktop-only import at CI time is cheap; debugging a runtime dart:io failure in a stakeholder's browser is not. The grep-based gating script and matching test prevent that.

7. Known Limits

- The app is a single-user, client-side scheduling tool. There is no shared state, multi-user collaboration, audit log, or user identity.
- Browser autosave is browser-profile-scoped. Switching browsers, profiles, or devices loses the autosave; admins must Save As to a portable bundle.
- showSaveFilePicker support varies by browser. On unsupported browsers (notably Firefox at the time of writing) the save flow falls back to a normal download.
- AWS deployment currently overwrites /var/www/html on EC2 in place. There is no automated rollback artifact; rollback is "redeploy a known-good commit" (see Operations_Runbook.md).
- Performance under unusually large datasets (well beyond the 24-course / 267-person fixture) is not separately profiled.

- The optional AI-assisted scheduling enhancement from D1 was deferred by stakeholder agreement and is not delivered.

8. Where to extend the system

Need	Touch this	Watch out for
Add a new export	New outputXyzToString() on Scheduling; route through exportTextFile()	Keep the gate check on StateOfProcessing so disabled actions stay disabled.
Add a new pipeline stage	New StateOfProcessing enum value + branch in _updateStateOfProcessing(); new compute module under lib/compute/	Update the bundled-state grammar and the legacy state loader (Scheduling.loadState).
Add CSV exports	Wrap the existing string-builders with a CSV serializer alongside the tab-delimited form	Mail-merge already produces tab-delimited rows; adding a CSV output is mostly a header + escaping pass.
Move off EC2 to S3 + CloudFront	Replace deploy.yml with aws s3 sync build/web s3:// … + an invalidation step	Update Operations_Runbook.md and the rotation table; archive the EC2 host config.
Add a backend (e.g., shared term schedules)	A new lib/api/ package with conditional Web/desktop transport; gate it the same way as I/O	Introduces auth and PII at rest — re-evaluate Security_Privacy_Accessibility_UX_Notes.md before implementing.

9. Repository inventory (for the next developer)

```
lib/
  scheduling.dart                 # facade — start here
  main.dart                       # Flutter entry + theme
  widgets/                        # UI
    screen.dart                   # main screen, file menus, autosave
    class_name_display.dart       # course rows + coordinator selection
    class_size_control.dart       # min/max class-size controls
    overview_data.dart            # overview/stats panel
    table/{main_table, drop_row, schedule_row, class_name_row, overview_row}.dart
    names_display_mode.dart, utils.dart
  compute/
    overview_data.dart            # placement + unmet wants
    course_control.dart           # drop, split mode, class size, coordinators
    schedule_control.dart         # 20-slot scheduling state
    split_control.dart            # split preview/commit (John Taber's algorithm)
    validate.dart                 # people vs courses consistency
  model/
    course.dart, person.dart, coordinators.dart, class_size.dart
    state_of_processing.dart, change.dart, exceptions.dart
  store/
    courses.dart, people.dart, list_item.dart
  io/
    text_file_store_factory.dart   # conditional import gate
    text_file_store_stub.dart, text_file_store_io.dart, text_file_store.dart
    web_download_factory.dart, web_download.dart, web_download_stub.dart, web_download_web.dart
    autosave_store_factory.dart, autosave_store_stub.dart, autosave_store_web.dart
    bundled_state.dart             # Save / Save As / Load bundle
    default_filename.dart           # timestamped export/save filenames
```

```
    export_text_file.dart          # unified save-as helper
    file_io_exceptions.dart
  platform/
    desktop_platform.dart, desktop_platform_stub.dart, desktop_platform_io.dart
  window_sizing/
    window_sizing.dart, window_sizing_desktop.dart, window_sizing_noop.dart
  theme.dart
test/                            # 21 .dart test files, 89 passing tests (see Testing_and_QA_Summary.md)
.github/workflows/
  test.yml                       # CI
  deploy.yml                     # build web → SCP to EC2
scripts/
  check_platform_gating.sh       # grep-based desktop-import gate
docs/                            # earlier-deliverable working docs
Project_Docs_D7/                 # D7 handoff source docs + rendered deliverables
```